

Module 12: Data Compression Techniques (Lossless & Lossy)

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DATA COMPRESSION: LOSSLESS vs LOSSY ARCHITECTURE

Run-Length Encoding, Huffman, Lempel-Ziv (LZ), Audio Psychoacoustics & Temporal Masking

LOSSLESS COMPRESSION (Zero Data Loss)

Principle: Exact bit-for-bit mathematical restoration. • **Mandatory For:** Text code (.c, .java), Databases, Executable binaries. • **1. Run-Length Encoding (RLE):** Replaces repeated streaks. "AAAAABBBCCDA" -> "A5B3C2D1A1" (12 bytes -> 10 bytes) • **2. Huffman Coding:** Variable length prefix codes based on freq. • **3. Lempel-Ziv (LZ77 / LZW):** Dictionary table index replaces words. Formats: ZIP, GZIP, PNG, 7z, FLAC, TIFF **LOSSY COMPRESSION (Perceptual Pruning)** • **Principle:** Discards imperceptible details for massive size cuts. • **Mandatory For:** Streaming Video (YouTube), Music, Photos. • **Audio Psychoacoustics & Masking (Slide 87 - 88):** • **Frequency Masking:** Loud tone drowns weaker tone at nearby Hz. • **Temporal Masking:** Loud strike (hammer) desensitizes ear briefly. • **Image Compression (JPEG):** Discrete Cosine Transform (DCT). Formats: MP3, AAC, JPEG, MPEG, H.264, MP4 **DATA COMPRESSION COMPARATIVE MATRIX & METRICS (SLIDE 84 - 88)** 1. Metrics & Equations • **Compression Ratio:** CR = Uncompressed Size / Compressed Size • **Space Savings:** Savings = 1 - (Compressed / Original) 2. Fidelity vs Compression Ratio • **Lossless Ratio:** Modest (usually 2:1 to 4:1). • **Lossy Ratio:** Extreme (10:1 to 50:1+). • Decompressing lossy video does not match original raw RGB sensor matrix bit-for-bit. 3. Psychoacoustic Masking • **Slide 88 Hammer Analogy:** When a loud hammer strikes, ear cannot detect a pin dropping immediately after. • Discarding masked sounds saves 90% space!

1. Data Compression Paradigms

Compression maximizes network transmission throughput and minimizes mass storage consumption by eliminating data redundancy.

Lossless vs Lossy Segmentation

Lossless: Strict bit-for-bit mathematical identity restoration (RLE, Huffman, Lempel-Ziv).
Lossy: Psychoacoustic and visual human perception threshold pruning (MP3, JPEG, H.264).

2. Lossless Compression Schemes

Algorithm	Operational Method	Target Data Stream
Run-Length Encoding (RLE)	Substitutes contiguous identical byte runs with character-count pairs	Scanned monochrome documents, fax bitmaps
Huffman Coding	Builds minimum redundancy binary prefix trees based on symbol probabilities	Variable-length character streams
Lempel-Ziv (LZW)	Maintains dynamic dictionary table substituting recurrent words with pointers	General archive utilities (ZIP, GZIP, PNG)

3. Psychoacoustic Audio Masking Mechanics

- **Frequency Masking:** A dominant louder sound tone suppresses softer tones residing within the critical spectral band.
- **Temporal Masking (Slide 88):** Loud auditory impulses render the human cochlea unresponsive for several milliseconds immediately preceding and following the transient event.